

Fleet Manager App - Minimum Viable Product (MVP)

Core features

- Drivers: CRUD with license expiry validation.
- Trucks: CRUD with inspection date validation.
- Jobs: CRUD with client, pickup, drop, dates, and status.
- Assignments: Link driver + truck + job, prevent conflicts.
- List views for jobs, drivers, trucks with status indicators.
- Simple dashboard: today's jobs, expiring licenses/inspections.
- Basic auth: Dispatcher (full) and Driver (read-only).
- Search & filters across jobs, drivers, trucks.
- Status flow: planned → in_progress → done.
- CSV export for Jobs, Drivers, Trucks.

Non-functional essentials

- Mobile-friendly pages (usable on phone).
- Audit basics: user + timestamp on edits.
- Backups: daily DB backup, keep 7 days.
- HTTPS enabled.
- Seed admin user with strong password.

Minimal data model

- Driver(name, phone, license_expiry, active)
- Truck(plate[unique], capacity_kg, inspection_due, active)
- Job(client, pickup, drop, start_date, end_date, status, notes)
- Assignment(job->Job, driver->Driver, truck->Truck, assigned_at)
- Optional: Setting(expiry_warn_days=30)

Validation rules

- Job dates invalid (start_date > end_date) → block save.
- Expired driver license → block assignment.
- Expired truck inspection → block assignment.
- Overlap: same driver/truck on overlapping jobs → block.

Minimum screens (URLs)

- /dashboard/
- /jobs/ (list, create, detail)
- /drivers/ (list, create, detail)
- /trucks/ (list, create, detail)
- /assignments/new?job=
- /exports/ (CSV)
- /login/

Build order

- Models + admin (instant CRUD).
- Jobs list + create form.
- Drivers & Trucks lists.
- Assignment form + validations.
- Dashboard.
- Driver read-only view.
- CSV export.

Done criteria

- Can create job, assign valid driver+truck, update status.
- Illegal assignments blocked (overlaps, expired docs).
- See upcoming expiries on dashboard.
- Driver can log in to see only their jobs.
- Can export last month's jobs to CSV.